

Master Learning Path – Basic → Advanced (~1 CR product companies)

MERN MASTERY · 1 CR PRODUCT PREP · HARSH VARDHAN GUPTA



Program dates: 17 Aug 2026 → 14 Feb 2027 (182 days)



Slots (IST): Mon–Fri 7:30–8:30pm & 9–10pm · Sat–Sun 9am–12pm, 3–5:30pm, 6–8:30pm



Reminders: Import <calendar/mern-mastery-study.ics> → harsh.gupta@getreelax.com

Rule: Do NOT skip phases. Each phase unlocks the next.

Goal: ~1 CR at product companies (Google L4–L5 / Meta E4–E5 / Amazon SDE2–3 / Microsoft / Uber / Atlassian). 30–80 LPA becomes a byproduct.

1 CR overlay: [20-1cr-product-company-complete.pdf](#) — every question type, company loops, product DSA/HLD, optional months 7–9.

Phase map (strict order)

PHASE 0 – Web & Environment (Days 0–3, optional if experienced)
↓
PHASE 1 – JavaScript Foundations (Days 1–21)
↓
PHASE 2 – TypeScript + Git (Days 22–28)
↓
PHASE 3 – Frontend (HTML/CSS recap → React → Next.js) (Days 29–56)
↓
PHASE 4 – Backend + Databases (Days 57–84)
↓
PHASE 5 – Production Projects (Days 85–112)
↓
PHASE 6 – DSA + LLD: Easy → Hard (Days 113–140) [see PDF #19 + #25]
6A Easy DSA → 6B Easy LLD → 6C Medium DSA → 6D Medium LLD
→ 6E DP/intervals/backtracking → 6F Company-tagged DSA → 6G Backend + frontend MC
↓
PHASE 7 – Product System Design (Days 141–168) [see PDF #19 + #25]
7A SD basics → 7B Easy HLD → 7C Feed/Notify → 7D Chat/Pay/Uber
→ 7E Distributed → 7F YouTube/Drive/Typeahead → 7G CS + FE MC + loops
↓
PHASE 8 – Interview execution (Days 169–182) [see PDF #19 + #25]
8A Review → 8B Partial mocks → 8C Combined mocks
→ 8D Company-shaped loops → 8E Apply + negotiate
↓
PHASE 9 – Optional months 7–9 (after Day 182) [see PDF #25 §11]
Hard DSA fluency → more product HLD → 2 loops/week → 1 CR pipeline

Complete PDF order (do not reorder)

Order	PDF	Phase	When (Days)	Prerequisites
0	00b-html-css-web-fundamentals-complete.pdf	0	Before Day 29	None
1	01-javascript-complete.pdf	1	1-21	None
2	02-typescript-complete.pdf	2	22-28	PDF #1
3	00-git-complete.pdf	2	22-28	PDF #1
4	08b-dsa-datastructures-fundamentals-complete.pdf	1-6	1-112 (parallel)	PDF #1 §arrays
5	03-react-complete.pdf	3	29-49	PDF #0, #2
6	04-nextjs-complete.pdf	3	50-56	PDF #5
7	05-nodejs-express-complete.pdf	4	57-63	PDF #2
8	07-auth-devops-complete.pdf	4	64-70	PDF #7
9	06-databases-complete.pdf	4	71-84	PDF #7
10	10b-networking-http-fundamentals-complete.pdf	4	78-84	PDF #7
11	– Projects –	5	85-112	PDF #5-10

12	08-dsa-patterns-complete.pdf	6	113-126	PDF #4 + 100 LC
13	09-lld-complete.pdf	6	119-140	PDF #2, #5
14	14-advanced-dsa-hard-complete.pdf	6	127-140	PDF #12
15	15-machine-coding-complete.pdf	6	118-140	PDF #13
16	10c-system-design-prerequisites-complete.pdf	7	141-143	PDF #10, #9
17	10-system-design-complete.pdf	7	141-154	PDF #16
18	11-distributed-systems-complete.pdf	7	145-158	PDF #16, #17
19	12-microservices-kafka-complete.pdf	7	151-162	PDF #18
20	13-kubernetes-cloud-advanced-complete.pdf	7	155-165	PDF #7 § Docker
21	16-behavioral-senior-complete.pdf	8	141-182	None
22	17-interview-qa-80lpa-master.pdf	8	Tiered – see below	Phase-appropriate
23	18-80lpa-roadmap-	All	Reference	Read after this doc

	addendum.pdf			
24	19-phases-678-easy-to-hard-complete.pdf	6-8	Days 113-182	Phase 6-8 master guide
25	20-1cr-product-company-complete.pdf	6-9	Days 113-182 + optional 183-273	1 CR question types, product DSA/HLD, company loops
26	21-1cr-remaining-gaps-complete.pdf	All	Day 1 (MERN verdict) + Month 4 OA/SQL + Days 165-171	OA, timed SQL, browser, a11y, GraphQL-lite, 1 Cr math
27	22-top-100-recruitment-processes-complete.pdf	8	Days 167-182 (and when a recruiter emails)	100 companies → process family → every hiring stage drill
28	23-top-product-companies-interview-100.pdf	All	2 questions/day from current topic	100 product-company questions with full answers

⚠ Phase 6-8: read PDF #19 for easy→hard sub-phases

Phases 6, 7, 8 each have 7 sub-phases from 🟢 Easy to 🔴 Hard.

Open 19-phases-678-easy-to-hard-complete.pdf before starting Day 113.

Interview Q&A study tiers (PDF #22)

Do NOT read all 140 Q&A on Day 1. Follow tiers:

Tier	When	Categories from PDF #22
Tier 1	Month 1 (Days 1-28)	JavaScript (15), TypeScript (10)
Tier 2	Month 2-3 (Days 29-84)	React (15), Node (15), Databases (15)
Tier 3	Month 4-5 (Days 85-140)	LLD (15) + DSA review questions
Tier 4	Month 6 (Days 141-168)	HLD (20), Distributed (15), Behavioral (15)
Tier 5	Offer stage	Salary (5)

Daily habit: 2 Q&A from current tier only, explain aloud. **Also:** 2 questions from the **100-question bank** at the end of the current theory PDF, plus 2 from **23-top-product-companies-interview-100.pdf**.

Phase details

PHASE 0 – Web fundamentals (Days 0-3, before React)

PDF: **00b-html-css-web-fundamentals-complete.pdf**

Topic	Why before React
HTML semantic tags	JSX compiles to HTML
CSS box model, flexbox	Tailwind/CSS modules
DOM API	React virtual DOM comparison
HTTP request in browser	fetch, CORS later

Skip if: You built multi-page websites before.

Exercise: Build static portfolio page (no JS framework).

PHASE 1 – JavaScript (Days 1-21)

PDF: `01-javascript-complete.pdf` (read in section order 1→21)

Parallel from Day 8: Start `08b-dsa-datastructures-fundamentals-complete.pdf` §1–4 (arrays, strings, hash map) while doing LC easy.

Week	JS sections	DSA fundamentals (parallel)	LC count
1	§1–5 variables, functions	§1 Arrays	5 easy
2	§6–9 scope, closures, this	§2 Linked List	5 easy
3	§10–16 async, promises	§3 Stack, Queue	5 easy

Gate to Phase 2: Explain event loop + closures without notes. 15+ LC easy done.

PHASE 2 – TypeScript + Git (Days 22-28)

PDFs: #2, #3

Gate: TS Catalog API compiles with `strict: true`. Git branch + merge done.

PHASE 3 – Frontend (Days 29-56)

Order matters:

1. PDF #0 (HTML/CSS) — Days 29–30 if skipped Phase 0
2. PDF #5 (React) — Days 29–49
3. PDF #6 (Next.js) — Days 50–56

Parallel: Continue PDF #4 §5–7 (trees, graphs) + LC to 60 total.

Gate: Deployed blog on Vercel. Can explain `useEffect` vs `useLayoutEffect`.

PHASE 4 – Backend (Days 57-84)

Strict order:

1. PDF #7 Node/Express (57–63)
2. PDF #8 Auth (64–70)
3. PDF #9 Databases (71–77)
4. PDF #10 Networking/HTTP (78–81) ← **before system design**
5. Finish databases + Redis (82–84)

Why networking here: You need HTTP, TCP, DNS before HLD or distributed systems.

Gate: Same API in Mongo AND PostgreSQL (or MySQL). Know PG vs MySQL tradeoffs. JWT refresh works. 100+ LC.

PHASE 5 – Projects (Days 85-112)

Build only after Phase 4 gate passed.

Project	Days	Applies
TaskFlow	85-98	Auth, Mongo, Redis, Socket.io, Docker
ShopLite	99-112	PostgreSQL, payments, tests, CI

Parallel: PDF #4 §8–10 (heaps, graphs). LC to 150.

PHASE 6 – DSA + LLD (Days 113-140) – Easy → Hard

Full guide: [19-phases-678-easy-to-hard-complete.pdf](#)

Sub	Days	Difficulty	Focus
6A	113-116	● Easy	DSA pattern refresh (two ptr, hash, stack)
6B	117-119	● Easy	LLD/OOP: Snake & Ladder, Vending Machine
6C	120-123	● Medium	DSA trees & graphs (BFS/DFS)
6D	124-127	● Medium	LLD: LRU Cache, Rate Limiter, Parking Lot
6E	128-132	● Med-Hard	DP, intervals, heap, backtracking
6F	133-135	● Hard	Company-tagged + hard LC + follow-ups
6G	136-140	● Hard	Backend MC + frontend autocomplete

Gate to Phase 7: 200 LC, 6 LLD, 1 backend MC + 1 frontend MC.

PHASE 7 – System Design (Days 141-168) – Easy → Hard

Full guide: [19-phases-678-easy-to-hard-complete.pdf](#)

Sub	Days	Difficulty	Focus
7A	141-143	● Easy	SD building blocks (10c: cache, LB, estimation)
7B	144-147	● Easy	Simple HLD: URL Shortener, Paste Bin
7C	148-152	● Medium	Feed, Notification, Instagram
7D	153-156	● Med-Hard	Chat, E-commerce, Payment, Uber
7E	157-160	● Hard	Distributed: CAP, Raft, Saga (PDF #11)
7F	161-164	● Hard	Product HLD: YouTube, Drive, Typeahead, seats
7G	165-168	● Hard	CS fundamentals + FE MC + company loop map

Gate to Phase 8: 10 product HLDs, CS pack, 8 tagged STAR stories.

PHASE 8 – Interview execution (Days 169-182) – Easy → Hard

Full guide: [19-phases-678-easy-to-hard-complete.pdf](#)

Sub	Days	Difficulty	Focus
8A	169-171	🟢 Easy	Weak area review, resume, GitHub polish
8B	172-174	🟡 Medium	Partial mocks (1 DSA, 1 LLD, 1 easy HLD)
8C	175-177	🟠 Med-Hard	Combined mocks (DSA + HLD back-to-back)
8D	178-180	🔴 Hard	Company-shaped loops (Google/Meta vs Amazon/Atlassian)
8E	181-182	🔴 Hard	Apply 8 product companies, 1 CR negotiate

Final gate: Pass full loop mock (2 DSA + 1 product HLD + behavioral) before mass applying.
If not 1 CR ready: Phase 9 (optional 90 days) in PDF #25 §11.

Common skip mistakes (avoid these)

Mistake	Why it fails	Fix
Reading distributed systems (PDF #18) before networking (PDF #10)	No HTTP/TCP context	Do PDF #10 in Month 3
Advanced DSA (PDF #14) before data structures (PDF #4)	Can't implement heap/trie	PDF #4 from Week 2
Interview Q&A Tier 4 in Month 1	Overwhelming, no context	Follow tier table
System design (PDF #17) before SD prerequisites (PDF #16)	Missing scaling basics	PDF #16 first
Skipping HTML/CSS before React	JSX/DOM confusion in interviews	PDF #0
Machine coding (PDF #15) before LLD basics (PDF #13)	Can't design classes	PDF #13 Week 17
Kafka (PDF #19) before distributed (PDF #18)	No CAP/consensus base	PDF #18 first

Weekly PDF checklist (print this)

Week	Theory PDFs (in order)	Exercise PDF	Interview Tier
1	01 §1-5	week-01	Tier 1 JS
2	01 §6-11, 08b §1-2	week-02	Tier 1 JS
3	01 §12-21, 08b §3	week-03	Tier 1 JS
4	02, 00-git	week-04	Tier 1 TS
5	00b, 03 §1-5	month-02 §W5	Tier 2 React start
6	03 §6-8	month-02 §W6	Tier 2 React
7	03 §9-12, 08b §5-6	month-02 §W7	Tier 2 React
8	04	month-02 §W8	Tier 2 React
9	05	month-03 §W9	Tier 2 Node
10	07 §1-4	month-03 §W10	Tier 2 Node
11	06 §Mongo	month-03 §W11	Tier 2 DB
12	06 §PG, 10b	month-03 §W12	Tier 2 DB
13-16	Projects + 07 §Docker	month-04	Tier 3 LLD
17	6A-6B	08 §1-4, 09 SOLID	month-05 §W17
18	6C-6D	08 §5-8, 09, 15 LRU	month-05 §W18
19	6E	08 §DP, intervals, backtracking	month-05 §W19
20	6F-6G	14 hard, 15 MC + FE autocomplete, 20-1cr §4	month-05 §W20
21	7A-7B	10c, 10 URL/Paste	month-06 §W21
22	7C-7D	10 Feed/Chat/Payment/Uber	month-06 §W22
23	7E-7F	11 + 10 §YouTube/Drive/Typeahe ad (20-1cr)	month-06 §W23

24	7G + 8A	20-1cr CS + FE MC + 16 behavioral	month-06 §W24
25	8B-8C	Review weak	month-06 §W25
26	8D-8E	Full mocks + apply	month-06 §W26

Success gates (must pass before next phase)

Gate	Requirement
→ Phase 2	FizzBuzz + closure + 15 LC easy
→ Phase 3	TS strict compiles + Git merge
→ Phase 4	React todo + hooks quiz
→ Phase 5	Auth API + both DBs
→ Phase 6	2 deployed projects
→ Phase 7	200 LC + 6 LLD + 1 FE MC
→ Phase 8	10 product HLD + CS pack + 8 tagged STAR
→ Offer	Full company-shaped loop mock pass

Where files live

```
pdf/
├── 00-LEARNING-PATH-COMPLETE.pdf    ← THIS FILE (read first)
├── 00-ROADMAP-COMPLETE.pdf          ← Daily tasks
├── theory/
│   ├── 00b-html-css-web-fundamentals-complete.pdf
│   ├── 08b-dsa-datastructures-fundamentals-complete.pdf
│   ├── 10b-networking-http-fundamentals-complete.pdf
│   ├── 10c-system-design-prerequisites-complete.pdf
│   └── ... (all others in order above)
└── exercises/
    └── ... (match weekly checklist)
```

Start: Read this PDF → then `00-ROADMAP-COMPLETE.pdf` Day 1.